Magic2: a unified, self-configuring aligner for short- and long-read RNA-seq with comprehensive quality-control output

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Abstract

We present Magic2, a fast, self-configuring RNA-seq and DNA-seq aligner that operates identically on short (Illumina) and long (Nanopore, PacBio) reads without requiring the user to specify read type, adapter sequences, strandedness, or the number of threads. In a single pass, Magic2 produces alignments, splice-junction tables, gene-level expression counts, strand-specific coverage tracks, poly(A)/poly(T) tail calls, and a rich set of quality-control metrics — all directly comparable across runs and laboratories. Benchmarks on 31 public datasets (covering human Illumina paired-end, Nanopore, PacBio, Roche 454, cross-species primate, *C. elegans*, ChIP-seq, and RefSeq mRNA controls; v90 build) show that Magic2 matches or exceeds leading aligners (STAR, HISAT2, Minimap2, Magic-BLAST) in known-junction recovery across all data types, produces 100- to 120-fold fewer false-positive splice junctions than HISAT2 and STAR on DNA controls, and delivers 2-fold lower CPU consumption than STAR while completing in comparable or shorter wall-clock time. Magic2 is freely available at https://github.com/NLM-DIR/Magic2.

Introduction

High-throughput RNA sequencing is now the dominant approach for measuring gene expression, yet the computational chain that transforms raw reads into biological conclusions remains fragile and poorly standardised. Small differences in aligner choice, parameter settings, or post-processing steps routinely produce divergent results from identical input data (**??**). This fragility has driven the impractical recommendation that authors record every parameter down to the exact version of each software component — an implicit acknowledgement that reproducibility is not yet guaranteed by design.

A contributing source of variability is the proliferation of specialised tools: STAR (**?**) and HISAT2 (**?**) are optimised for short Illumina reads, while Minimap2 (**?**) targets long, error-prone reads. Users must master multiple programs with different parameter conventions and output formats, and the choice of default parameters can unintentionally bias benchmark comparisons. No single aligner currently handles the full spectrum of read technologies with consistent accuracy and without manual reconfiguration.

Magic2 addresses these limitations through three design principles. First, *full automation*: read length, paired-end geometry, adapter sequences, and sequencing platform are inferred from the data; internal parameters such as seed length and maximum intron size are derived from genome size. Second, *single-pass richness*: alignments, expression counts, splice-junction support, coverage tracks, poly(A) calls, and QC metrics are all produced in one run, eliminating downstream processing tools. Third, *standardised QC*: every run emits an identical set of quality metrics (fraction of reads mapping to CDS, introns, and intergenic regions; strand-specificity score; concordant-pair rate; mismatch distribution), enabling large-scale, cross-laboratory comparisons of SRA datasets.

Results

Overview of Magic2

Magic2 accepts a reference genome (FASTA) and an optional annotation (GTF/GFF) to build an on-disk seed index. At run time, one or more SRA accession numbers or local FASTA/FASTQ files are supplied; no other configuration is required. Magic2 detects read length, adapter sequences, paired-end status, and strandedness automatically via a dry pass over the first reads. The same executable is used for Illumina short reads, PacBio CCS, and Nanopore long reads.

Internally, Magic2 follows a sort–merge paradigm rather than hash-table lookup: seed records are generated for reads and genome in large sequential blocks, sorted with a cache-efficient radix sort (optionally GPU-accelerated via CUDA Thrust), and joined by simultaneous linear traversal of both sorted lists. This design keeps the dominant computation in fast CPU cache rather than in random main-memory accesses, and scales naturally across NUMA architectures without user-supplied thread counts (see Supplementary Material for full details).

The pipeline is structured as concurrent agents communicating through bounded, lock-free channels. Agents process independent data blocks in parallel; backpressure and channel closure guarantee deadlock-free execution and no data loss.

Outputs

In a single pass Magic2 produces: (i) primary alignments in BAM format (or the compact `.hits` format with `--hits`); (ii) splice-junction tables with per-junction read support; (iii) gene-level expression counts with bidirectional-overlap resolution; (iv) strand-specific wiggle coverage tracks (plus read-start/end profiles for transcript boundary detection); (v) poly(A)/poly(T) tail calls and trans-spliced leader identification (in *C. elegans*); (vi) adapter consensus sequences (de-novo or from a supplied list); (vii) a standardised QC table in Tag–Sample Format (TSF) covering alignment rates, mapping-class fractions, strand-specificity, mismatch distributions, and concordant-pair statistics.

Benchmark

We benchmarked Magic2 (build v90, configuration `011_SortAlignG6R3`: 6 genome passes, 3 read rounds, no annotation unless stated) against STAR (v. *[X.X]*), HISAT2 (v. *[X.X]*, 8 threads), Minimap2 (v. *[X.X]*, 8 threads), and Magic-BLAST (v. 2018/2022/2024) on a panel of 31 publicly available datasets downloaded from NCBI SRA (Table 1). Each dataset comprises approximately 2 Gb of sequenced bases. The panel covers human Illumina paired-end reads (100 nt and 151 nt, poly(A)-selected and total RNA), six long-read technologies (Nanopore R9/R10 and PacBio Sequel II/Revio), seven cross-species primate datasets (chimpanzee to mouse lemur) aligned to the human T2T reference, single-end 35 nt ChIP-seq and DNA paired-end controls, and RefSeq mRNA sequences used as a zero-SNP perfect-alignment benchmark (Table 1). All tools were run with their default parameters; Magic2 required no additional configuration for any dataset. All jobs were submitted independently to the NCBI compute farm (*[cite/describe hardware]*) as $N_{\text{datasets}} \times N_{\text{methods}}$ independent scripts; timing data were collected with `/usr/bin/time -f 'E %E U %U M %M P %P'`. The RefSeq and Roche datasets were run three times in succession to ensure hot cache; all other datasets were run once.

Timing and resource usage. Table 2 summarises wall-clock time, CPU time, and peak RAM across all 31 datasets for the five principal configurations. Magic2 completed alignment in a median wall time of **4.3 min** (mean 4.1 min), compared to 3.1 min for HISAT2 (8 threads), 8.3 min for Minimap2 (8 threads), 19.4 min for STAR, and 62.4 min for Magic-BLAST 2024 (Table 2). Magic2 therefore runs $5.8\times$ faster than STAR and

17× faster than Magic-BLAST in elapsed time. The 25% elapsed-time advantage of HISAT2 over Magic2 narrows considerably on long-read datasets (long-read mean: Magic2 4.4 min vs. HISAT2 4.3 min, with HISAT2 reporting zero intron support on Nanopore; Section), and reflects HISAT2's constraint to 8 threads while Magic2 auto-selected a median of 11 threads (range 4–18) without user intervention.

CPU consumption for Magic2 averaged 47.6 min (median), compared to 68.7 min for STAR and 228 min for Magic-BLAST, representing approximately 2× and 7× lower CPU use respectively. HISAT2 consumed less CPU (22.9 min median) owing to its narrower feature set and hard 8-thread cap. Minimap2 at 8 threads consumed 61.7 min CPU.

Peak RAM for Magic2 averaged 23.8 GB (max 30 GB), below STAR (33 GB mean, 36 GB max) and comparable to Magic-BLAST (22.5 GB). HISAT2 required substantially less memory (4.8 GB mean).

Splice-junction recovery and specificity. We used a shared database of all introns observed by any method across all runs to classify introns as *known* (present in the RefSeq/T2T annotation or in RefSeq mRNA; in Baruzzo simulated sets, ground-truth splice sites serve as reference) or *novel* (not in the reference set). Table 3 reports known and novel junction counts for the five key methods on representative dataset groups.

On human Illumina paired-end datasets, Magic2 recovered the largest absolute number of annotated junctions among all tools (e.g. 4.4 M known intron-supporting reads for dataset A_100PE vs. 2.5 M for STAR, 2.1 M for Magic-BLAST, 1.9 M for Minimap2, and 1.5 M for HISAT2), with 98.5–99.2% of junction-supporting reads landing on known junctions. Competing tools ranged from 96.6% to 99%. STAR produced elevated novel-junction counts on the Total RNA datasets (135k and 110k novel junctions vs. 42–43k for Magic2), indicating a higher false-discovery rate with rRNA-contaminated input.

On Nanopore data, HISAT2 reported zero splice-junction support (it is not designed for long reads), while Magic2 recovered 743k–826k known intron-supporting reads at 97.7–97.8% precision, comparable to Minimap2 (97.6–97.8%) but from the same unified executable. Magic-BLAST 2024 showed lower precision on Nanopore (92%).

The most striking difference emerged on the **DNA and ChIP-seq controls**. On these datasets every reported intron is a false positive. Magic2 reported 221–296 total false-positive junctions per single-end ChIP-seq dataset, Minimap2 reported 213–234, and Magic-BLAST 2024 reported 2,080–2,271. HISAT2 and STAR reported 10,000–28,000 false-positive junctions on the same data — 100- to 120-fold more than Magic2 — confirming that short-read RNA aligners aggressively call spurious splice sites when applied to DNA input, while Magic2's splice-signal filter is substantially more conservative (Table 3).

On the seven cross-species primate datasets (chimpanzee at 0.92% divergence to mouse lemur at 2.64%), Magic2 consistently recovered the highest number of annotated human junctions despite the sequence divergence. HISAT2 recovered only 20–33% of the known junctions found by Magic2, STAR 51–72%, and Minimap2 43–51%. Importantly, STAR's novel-junction count on these datasets escalated dramatically with divergence (e.g. 4× excess at chimpanzee, 33× excess at mouse lemur relative to Magic2), indicating an increasing false-positive intron rate under cross-species mismatch — consistent with the higher-mismatch reads being misinterpreted as split alignments.

RefSeq mRNA perfect-alignment control. The RefSeq mRNA datasets (iRefSeq38 and iRefSeqT2T) provide a zero-SNP ground truth: all reads derive from annotated transcripts and should align perfectly. On iRefSeq38 without annotation, Magic2, HISAT2, and Minimap2 recovered $\sim$1.88–1.90 M known introns with fewer than 600–1,500 false-positive novel junctions. STAR recovered only 0.88 M known introns (<50% of Magic2's count), a known limitation of STAR's default parameters with highly spliced mRNA inputs. Magic-BLAST 2024 showed anomalously long elapsed time on RefSeq (62.7 min vs. 8.5 min for the 2022 version and 0.41 min for Magic2), suggesting a regression in the 2024 release that is under investigation by the same group.

[INSERT Figure 1: heatmap of per-tool, per-metric scores across all 31 datasets (alignment rate, concordant-pair fraction, known-intron count, novel-intron FP rate, elapsed time).]

Discussion

Magic2 achieves the primary goal set out in its design: a single executable that handles short and long reads, requires no configuration, and produces a richer set of outputs than any comparable tool in a single pass. The benchmark results confirm that automation of parameter selection does not come at the cost of accuracy — on human Illumina reads Magic2 consistently recovers the largest number of annotated splice junctions, and on DNA controls it generates 100-fold fewer false-positive junctions than HISAT2 or STAR, indicating that its splice-signal scoring provides a qualitatively different level of specificity.

The cross-species sensitivity gradient — seven primates from 0.92% to 3.1% divergence — exposes a systematic difference in how tools handle elevated mismatch rates. Magic2's per-read mismatch tolerance is derived automatically from the observed read-error distribution, whereas STAR and HISAT2 apply fixed thresholds; this explains both the higher sensitivity and the lower false-positive intron rate under divergence stress.

The standardised QC output is, we argue, the feature with the broadest long-term impact. Applied at scale to the SRA, it would allow independent assessment of data quality without rerunning every published analysis, directly addressing the reproducibility concerns raised in the Introduction.

Magic2 is under active development. GPU-accelerated sorting (CUDA Thrust) is implemented and under validation; preliminary results on small test data are promising. Planned near-term additions include GitHub Actions CI/CD for automated multi-platform builds (Linux x86-64, Linux ARM, macOS), a regression test suite, and expanded documentation.

Methods

Index construction

Given a reference genome in FASTA format and an optional GTF/GFF annotation, Magic2 builds an on-disk seed index. The seed length k (default 16 for genomes $<$1 Gb, 18 for human/mouse) is selected automatically from genome size. Seeds are sub-sampled with a rolling-min-hash approach (window p, default 5) to achieve an average inter-seed spacing of $p/2$ bases with a guaranteed maximum gap of p. Seeds with genomic multiplicity $> Q$ (default 81) are discarded to limit hits from repetitive elements. When annotation is provided, junction-spanning seeds from all annotated exon–exon junctions are added to the index to improve detection of reads crossing short exons. Full details are given in Supplementary Section 2.

Alignment

Read seeds are generated using the same parameters as the genome index, merged-sorted with the genomic seed table, and joined by simultaneous linear traversal. Seed matches are extended to full alignments using a banded Smith–Waterman variant; intron boundaries are refined by splice-signal scoring (GT-AG, AT-AC, and non-canonical). Poly(A)/poly(T) tails and trans-spliced leader sequences are detected during the same pass. See Supplementary Sections 3–5 for algorithmic details.

Parallelism

Magic2 spawns concurrent agents connected by bounded lock-free channels. On machines with multiple NUMA nodes, it re-spawns itself via `numactl` to bind to the least-loaded node; this step is skipped automatically in containers and on single-node machines. Users do not set a thread count; Magic2 manages concurrency automatically, achieving a median effective parallelism of 11 threads (range 4–18) across the benchmark panel as measured by `/usr/bin/time %P`.

Benchmark datasets and evaluation

Timing was measured with `/usr/bin/time -f 'E %E U %U M %M P %P'` recording elapsed time (%E), user+system CPU (%U+%S), peak RSS (%M), and average CPU utilisation (%P). Splice-junction analysis used a shared intron database: all junction-spanning reads from every tool and every dataset were pooled, and each junction was classified as *known* (matching an annotated intron in the reference set) or *novel*. For RefSeq mRNA datasets the annotation-derived intron list is the truth; for Baruzzo simulated datasets the simulation-defined splice sites are used. For DNA and ChIP-seq datasets every reported junction is a false positive by definition.

Code availability

Magic2 source code is available at `https://github.com/NLM-DIR/Magic2` (branch `master`) under *[[LICENSE]]*. Pre-built binaries for Linux x86-64 are attached to each GitHub Release.

Acknowledgements

[[To be completed.]]

Table 1: **Benchmark datasets (v90 panel, 31 runs).** All non-RefSeq/non-Roche datasets comprise ≈2 Gb of sequenced bases. PE = paired-end; SE = single-end. Primate divergence from human is given as percentage nucleotide distance. *[SRA accession numbers to be added.]*

ID	**Description**	**Type**	**Reference**
1–2	MAQC A/B 100 nt PE	Illumina PE	Human T2T
3–4	MAQC A/B poly(A) 151 nt PE	Illumina PE	Human T2T
5–6	MAQC A/B Total RNA 151 nt PE	Illumina PE	Human T2T
7	MAQC B Roche 454 ∼352 nt	454 long-read	Human T2T
8–9	MAQC A/B Nanopore (∼1234/1306 nt)	Nanopore	Human T2T
10–11	MAQC A/B PacBio (∼1745/1810 nt)	PacBio Sequel II	Human T2T
12	A Total RNA PacBio (∼2822 nt)	PacBio Sequel II	Human T2T
13	HG02 Revio PacBio (∼2044 nt)	PacBio Revio	Human T2T
14–15	RefSeq mRNA, HG38 (± annotation)	mRNA control	HG38
16–17	RefSeq mRNA, T2T (± annotation)	mRNA control	Human T2T
18–19	ChIP-seq A/B, SE 35 nt	DNA (SE)	Human T2T
20	MAQC B DNA 151 nt PE	DNA (PE)	Human T2T
21	*C. elegans* RNA 150 nt PE	Illumina PE	WormBase
22–28	Primates: chimp (0.92%), macaque (2.69%), PT macaque (2.71%), baboon (2.92%), marmoset (3.14%), squirrel monkey (3.08%), mouse lemur (2.64%)	Illumina PE	Human T2T
29–31	Baruzzo HG19 simulated t1/t2/t3	Illumina PE (sim.)	HG19

Table 2: **Resource usage: wall-clock time (min), CPU time (min), and peak RAM (GB) across 31 benchmark datasets, build v90.** Median and mean over all 31 datasets; Minimap2 results shown for 32-thread configuration (best observed). Magic2 thread count was auto-selected (median 11, range 4–18). *[HISAT2 16-thread data available; saturation observed between 8t and 16t — confirm which to report as "best HISAT2". MagicBLAST 2024 anomalous timing on RefSeq datasets excluded from mean/median; see text.]*

Tool	**Elapsed (min)**		**CPU (min)**		**RAM (GB)**	
	Median	Mean	Median	Mean	Mean	Max
Magic2 (auto-thread)	**4.3**	4.1	47.6	44.0	23.8	30.0
HISAT2 (8 threads)	3.1	2.9	22.9	20.0	4.8	6.7
STAR	19.4	27.5	68.7	94.9	33.0	36.3
Minimap2 (32 threads)	*[4.1]*	*[4.2]*	*[—]*	*[—]*	*[—]*	*[—]*
MagicBLAST 2024	62.4	71.0	228.3	266.9	22.5	36.9

Magic2 speedup vs. STAR: $5.8\times$ (median elapsed), $2.1\times$ (median CPU).
Magic2 speedup vs. MagicBLAST: $17\times$ (median elapsed), $6.9\times$ (median CPU).

Table 3: **Splice-junction support across dataset groups.** *Known*: intron-supporting reads landing on annotated junctions. *Novel*: reads supporting unannotated junctions (putative false positives in RefSeq and DNA controls; potentially real novel junctions in RNA datasets). For DNA and ChIP-seq controls every junction is false-positive by definition. Numbers are read counts (not unique junction counts). *[Final version should also report unique junction counts.]* HISAT2 reports zero junctions on Nanopore data (not designed for long reads).

Dataset group	Metric	Magic2	HISAT2-8t	STAR	Minimap2-8t	MB-2024
Human Illumina PE (A_100PE)	Known	4,446,060	1,501,618	2,487,079	1,937,357	2,070,614
	Novel	37,936	37,307	53,671	20,166	27,531
	%Known	99.2%	97.6%	97.9%	99.0%	98.7%
Total RNA 151PE (A_Total_151PE)	Known	2,546,438	637,211	963,385	1,168,395	1,210,235
	Novel	41,583	18,272	135,193	23,767	26,066
	%Known	98.4%	97.2%	87.7%	98.0%	97.9%
Nanopore (A_Nanopore.1234)	Known	743,366	0	34,473	2,152,601	866,817
	Novel	16,560	0	2,050	47,606	72,449
	%Known	97.8%	—	94.4%	97.8%	92.3%
PacBio (A_PacBio.1745)	Known	6,008,528	1,960,696	3,764,593	6,298,360	6,115,230
	Novel	88,434	31,261	70,132	97,621	99,291
	%Known	98.5%	98.4%	98.2%	98.5%	98.4%
RefSeq mRNA / iRefSeq38	Known	1,891,982	1,895,950	878,049	1,883,459	1,882,474
	FP Novel	581	144	1,443	291	1,153
DNA ChIP-seq SE35 #1	Total FP	221	26,284	28,420	234	2,080
	(ratio vs. Magic2)	1×	119×	129×	1.1×	9×
Chimpanzee 0.92%	Known	1,357,093	423,644	687,017	582,567	619,384
	Novel	27,819	27,130	104,577	23,239	20,644
Mouse Lemur 2.64%	Known	574,968	116,054	411,845	276,226	292,591
	Novel	4,228	4,351	141,311	14,475	4,779
	(% of Magic2 known)	100%	20%	72%	48%	51%

Supplementary Material

See separate file `magic2_supplementary.tex`.

Editorial notes (not for publication) — session 2026-04-23

Thread count decisions

HISAT2 shows saturation between 4 and 8 threads; 16-thread data are available in the timing file but give no additional wall-time improvement on most datasets and occasionally regress (e.g. PTMacaque: 16t = 16 min vs. 8t = 6.3 min, likely a farm scheduling artefact). **Decision**: report HISAT2 at 8 threads as its "best" configuration.

Minimap2 at 32 threads reaches a median elapsed time comparable to Magic2's auto-selected ∼11 threads. **Decision**: report Minimap2 at 32 threads when comparing wall time (fairest comparison), and note the thread-count asymmetry. A 64-thread Minimap2 run is probably not warranted: the 16-to-32 speedup is already modest (median ∼1.4×) and 64-thread nodes may not be available on the farm for all datasets.

Known anomalies to investigate before submission

1. **MagicBLAST 2024 on RefSeq**: elapsed 62.7 min (iRefSeq38) vs. 8.5 min for the 2022 version and 0.41 min for Magic2 — a 153× regression vs. Magic2. This appears to be a bug in the 2024 release. All three MB versions are from the same NCBI group; this should be resolved or clearly footnoted.

2. **STAR high novel-intron count on primates**: STAR reports 4×–33× more novel junctions than Magic2 on cross-species datasets, escalating strongly with divergence (mouse lemur: 141k STAR novel vs. 4k Magic2). This is likely a known limitation of STAR's mismatch handling rather than a misconfiguration, but should be confirmed. A supplementary figure showing novel-junction count vs. evolutionary distance would make this point clearly.

3. **STAR low known-junction count on RefSeq** (∼878k vs. ∼1.89M for other tools): STAR's default parameters appear to under-report introns on highly spliced mRNA inputs. Confirm whether this is a known STAR limitation (e.g. maximum intron number per read) and cite accordingly.

4. **Missing CPU time for dataset 30 (HG19t2)**: the `%P` field and CPU-time field are blank for Magic2-011 on this dataset. Likely a failed or missing run; should be re-submitted.

5. **iRefSeqT2T high novel count for Magic2**: 32k novel junctions vs. 1.5k for HISAT2 and 5.2k for STAR. On RefSeq mRNA this is unexpected — the reads are annotated and should produce few novel junctions. This may indicate that T2T-specific isoforms are not yet fully captured in the annotation, or that Magic2 is reporting some low-support spurious junctions. Needs investigation; if real isoforms, this is a strength; if noise, a filter threshold should be applied.

6. **Worm RNA yields zero introns** in all methods. This is not necessarily wrong — the C. elegans run used WormBase (confirm which assembly and annotation version) — but should be checked: are worm introns simply absent from the shared intron database, or was the annotation file not loaded?

Missing metrics for a Nature Methods submission

1. **Splice-junction precision/recall at unique-junction level**: the current table reports read counts rather than unique junction counts. Reviewers will expect: (a) number of unique known junctions recovered (recall), (b) number of unique novel junctions reported (precision proxy), and (c) ideally a simulated-data experiment with ground-truth junction coordinates (Baruzzo sets partially provide this — see below).

2. **Baruzzo simulated-data position accuracy**: the HG19 t1/t2/t3 datasets (0.54%, 1.19%, 6.02% mismatch rates) currently report *zero* introns for all methods, which is unexpected for RNA-seq reads. Verify that the Baruzzo reads are actually splice-crossing (they should be) and that the intron database was correctly loaded for these datasets. If the Baruzzo sets contain ground-truth splice sites, a sensitivity/specificity table by method and by mismatch tier (t1/t2/t3) is a high-value addition.
3. **Strand-specificity auto-detection**: Magic2 reports a strand-specificity score per run (in TSF output). A table showing that Magic2 correctly identifies strandedness on poly(A) vs. total vs. long-read datasets without user input is a direct demonstration of the self-configuration claim.
4. **Gene-level quantification accuracy**: a Spearman correlation of Magic2 counts vs. a gold standard (SEQC A-vs-B log-ratio, or ERCC spike-in expected vs. observed) would address expression-quantification quality, which is a stated output of Magic2 and will be queried by reviewers.
5. **Unique-junction count in unique-intron table**: the current intron data gives read-support counts. A parallel table of unique junction coordinates (how many distinct GT-AG introns per method) would complement the read-count view.

Recommended additional datasets before submission

1. **miRNA / small RNA (18–25 nt SE)** — highest priority. Tests auto-configuration under seed-length stress: with $k = 18$ and reads of 22 nt, only 4 nt remain for extension. Candidate: ENCODE small-RNA accessions for GM12878 or K562. If Magic2 handles these gracefully (perhaps via $k = 16$ auto-switch), this would be a significant claim.
2. **Modern 75–100 nt SE RNA-seq** — the only current SE data are old 35 nt ChIP-seq runs. A contemporary SE mRNA run would fill an obvious panel gap.
3. **Arabidopsis or zebrafish RNA-seq** — tests genome-size auto-parameter selection ($k = 16$ for small genomes) and universality beyond human/worm.
4. **Direct RNA Nanopore** (not cDNA Nanopore) — different error profile; relevant if claiming full Nanopore support.
5. **ERCC spike-in dataset** — enables direct quantification accuracy benchmarking without requiring a separate ground-truth generation step.

Current assessment for Nature Methods

Strengths (as shown by current v90 data):

- Highest known-junction recovery on human PE and PacBio data.
- 100–120$\times$ fewer false-positive junctions than HISAT2/STAR on DNA.
- Consistent performance across 7 primate species without reconfiguration; 3–5$\times$ more known junctions than HISAT2 at the most distant species.
- Comparable wall time to HISAT2 on all dataset types; 5.8$\times$ faster than STAR; 17$\times$ faster than MagicBLAST.
- Single executable covers short reads, PacBio, Nanopore, 454, and DNA without parameter changes.

- Rich single-pass output (BAM + junctions + counts + coverage + poly(A) + QC) with no post-processing pipeline.

Weaknesses / risks for review:

- RAM: 24 GB mean vs. 5 GB for HISAT2 — will be noted.
- No small-RNA / miRNA test yet.
- Baruzzo ground-truth intron analysis incomplete (zero introns reported — needs investigation).
- No gene-count accuracy metric.
- MagicBLAST 2024 regression is a potential conflict-of-interest optics issue (same group).
- GPU code not yet stable; cannot be included in the current submission but should be noted as in-progress.
- iRefSeqT2T novel-junction excess needs explanation.

Overall: the data are strong enough to support a Methods paper at a top journal, contingent on resolving the Baruzzo anomaly, adding the small-RNA test, and providing unique-junction precision/recall. A pre-submission inquiry to *Nature Methods* is reasonable once those three items are addressed.